Kelipatan Persekutuan Terkecil (KPK) & Faktor Persekutu

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Student

Student: Kheyla

Topic: Kelipatan Persekutuan Terkecil (KPK) & Faktor Persekutuan Terbesar (FPB)

Status: completed

Lesson Plan

Learning Objectives

- Calculate the Prime Factorization of two numbers using the factor tree method.
- Solve for the Greatest Common Factor (FPB) and Least Common Multiple (KPK) of two numbers using their prime factors.
- Apply the concept of KPK to determine the timing of recurring events in real-world scenarios.
- Apply the concept of FPB to determine the maximum number of equal groups in distribution scenarios.

Lesson Information

Lesson Sections

1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Kheyla! I am so excited to see you today. Before we jump into our new mission, let's flex our brain muscles with a quick 'Number Power Up'! I'm going to share my screen. Look at these two challenges. First, for our place value check: Can you tell me the value of the digit '7' in 4,725? Great! Now, if I have 2 thousands blocks and 5 tens blocks, what number do I have? Perfect. Now, let's look at our diagnostic game: If I have the numbers 12, 18, and 24, which one is NOT a multiple of 6? And finally, what is the largest prime number less than 10? You nailed it!

Activities:

- *Place Value Visual Challenge: Identifying values in 4-digit numbers.*
- *Building Numbers: Interpreting virtual base-ten blocks (Thousands/Tens).*
- *Multiple Hunter: Identifying multiples within a set.*
- *Prime Quick-Fire: Identifying small prime numbers (2, 3, 5, 7).*

2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Imagine you are an event organizer, Kheyla! You have 12 chocolate bars and 18 vanilla cupcakes. You want to make snack bags for your friends. Every bag must be identical, and you don't want any leftovers. To be the best organizer, you need two secret tools: FPB (to split things up equally) and KPK (to find out when events happen at the same time). Ready to become a Master Organizer?

Activities:

- *Storytelling session: The Snack Bag Dilemma.*

3. Review of Prior Knowledge (5 minutes)

Teacher Script:

To use our secret tools, we need to know how to build 'Factor Trees.' Remember, we keep splitting a number until we only have Prime Numbers—those are our 'seeds.' Let's try splitting the number 12 together on the whiteboard. 12 is 2 times what? Correct, 6. And 6 is 2 times... 3! So 12 is 2 x 2 x 3.

Activities:

- *Interactive Factor Tree drawing for 12 and 18.*

4. Problem Introduction (10 minutes)

Teacher Script:

Now, let's find the FPB and KPK of 12 and 18. Look at our factors for 12: (2^2 x 3). And for 18: (2 x 3^2). Here is the secret rule: For FPB, we only take the 'Common' factors with the *smallest* power. They are like pieces that fit in both boxes. For KPK, we are greedy! We take *every* factor we see, and if they repeat, we take the *biggest* power.

Activities:

- *Color-coding factors: Circle common factors in red for FPB.*
- *Highlighting all factors in yellow for KPK.*

5. Guided Practice (15 minutes)

Teacher Script:

Let's solve our Organizer Problem. We had 12 chocolates and 18 cupcakes. To find the most bags we can make, we need the FPB. We found the factors: 12 = 2x2x3 and 18 = 2x3x3. The common ones are one '2' and one '3'. So FPB = 2 x 3 = 6 bags! Now, imagine a neon light flashes every 12 seconds and a bell rings every 18 seconds. When do they happen together? Use KPK! We take 2^2 x 3^2 = 4 x 9 = 36. They match at 36 seconds!

Activities:

- *Solving the Snack Bag problem (FPB).*
- *Solving the Light/Bell sync problem (KPK).*

6. Independent Practice (15 minutes)

Teacher Script:

Your turn, Master Organizer! I've put two tasks on the screen. Task 1: Find the FPB and KPK of 20 and 30. Task 2: A bus departs every 15 minutes, and another every 20 minutes. If they leave together at 08:00, when is the next time? Type your answers in the chat when you're done!

Activities:

- *Problem 1: Calculation of FPB/KPK for 20 and 30.*
- *Problem 2: Word problem involving bus schedules (KPK).*

7. Consolidation & Reflection (5 minutes)

Teacher Script:

You did it! Today you learned how to use Factor Trees to find FPB (for sharing) and KPK (for timing). What was the trickiest part? You handled it like a pro. Don't forget to practice! For your homework and next lesson, please join https://learn.algonova.id/login and head to your dashboard. See you next time!

Activities:

- *Reflective summary.*
- *System login demonstration.*

Homework

Medium Questions:

1. Find the FPB of 24 and 36.
2. Find the KPK of 10 and 15.
3. Amira has 15 red ribbons and 20 blue ribbons. What is the maximum number of friends she can give an equal set of ribbons to?
4. A lighthouse flashes every 8 seconds and another every 12 seconds. After how many seconds will they flash together?

Hard Questions:

1. Determine the KPK and FPB of 45 and 60 using the factor tree method.
2. Three bells ring at intervals of 12, 18, and 30 minutes. If they ring together now, in how many minutes will they ring together again?
3. A teacher has 48 pencils and 64 erasers. She wants to make goody bags with the same number of items in each. What is the largest number of bags she can make?
4. Find a number that is a multiple of both 14 and 21, but is also greater than 50.
5. REVIEW: Nilai Tempat Hingga Ribuan - What is the value of 5 in 5,201 and 1,520? Calculate the difference between these two values.
6. REVIEW: Diagnostik Interaktif - List all prime factors of 100 and determine if the sum of these factors is a prime number.

Answer Key:

1. FPB(24,36): 24=2^3x3, 36=2^2x3^2. FPB=2^2x3=12.
2. KPK(10,15): 10=2x5, 15=3x5. KPK=2x3x5=30.
3. FPB(15,20)=5 friends.
4. KPK(8,12)=24 seconds.
5. 45=3^2x5, 60=2^2x3x5. FPB=3x5=15; KPK=2^2x3^2x5=180.
6. KPK(12,18,30)=180 minutes.
7. FPB(48,64)=16 bags.
8. KPK(14,21)=42. Next multiple is 84. Answer: 84.
9. REVIEW: 5,000 and 500. Difference = 4,500.
10. REVIEW: 100=2^2x5^2. Prime factors are 2 and 5. Sum=7. 7 is prime (Yes).

REVIEW: Nilai Tempat Hingga Ribuan - Represent 3,042 using words and explain why there is no 'hundreds' mentioned.
REVIEW: Nilai Tempat Hingga Ribuan - If you move the digit '9' from the tens place to the thousands place in the number 1,290, what is the new number and how much larger is it?
REVIEW: Diagnostik Interaktif - Which of these numbers has the most factors: 12, 15, or 17? Explain why.
REVIEW: Diagnostik Interaktif - Is the number 1,234 divisible by 2, 3, or 5? Use divisibility rules.

Parent Reminder

Today's Math Milestone: KPK & FPB

Hi! Today Kheyla learned how to solve 'Master Organizer' problems using KPK and FPB. She practiced using factor trees to split groups equally and find matching cycles. Great focus today! Please help her log in to https://learn.algonova.id/login to finish her practice.